Ex No: 10

27/10/2026

# Interworking with Routing in Cisco Packet Tracer Simulator

A) Design and Configure a simple Internetwork Router in Cisco Packet Tracer

Aim: To design and configure two different networks using a router and switches, assign static IP addresses and verify Packet communication between the networks

Steps:

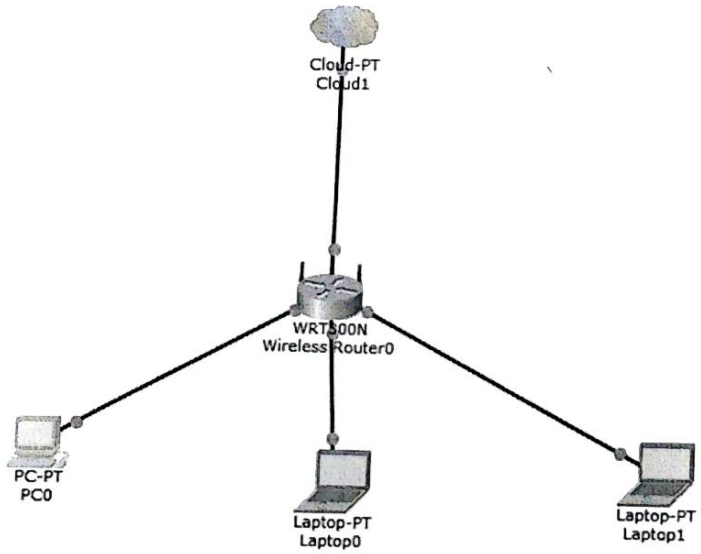
1. Open Cisco Packet Tracer
2. Add 1 router, 2 switches and 3-4 PCs.
3. Connect PCs to switches using copper straight-through cables
4. Connect both switches to the router
5. Assign static IP addresses to all PC's
6. Configure the router interface with suitable IP address
7. Enable the router interface using no shutdown.
8. Set the router interface IP as the default gateway for each network.
9. Use the ping command to check connectivity.
10. Select Simulation mode and observe the packet transmission path through the router.

Result:

The two networks were successfully connected using a router and communication between different networks was verified successfully using the ping command and Simulation mode.

```
Router#enable
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.10.1    YES manual up                    up
GigabitEthernet0/1     192.168.20.1    YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#
```

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```
Router(config)#interface gigabitethernet 0/1
Router(config-if)#ip address 192.168.20.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state
to up
exit
```

b) Internetwork using Wireless Router, NDHCP Server and Internet Cloud.

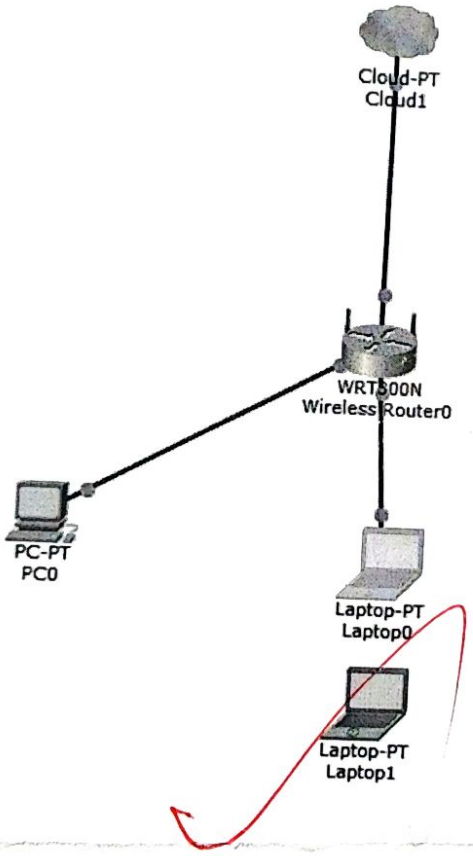
AIM: To design and configure a wireless internetwork using a wireless router, DHCP server and internet cloud and verify automatic IP address allocation & connectivity.

STEPS:

1. Open Cisco Packet Tracer
2. Add a wireless Router, Laptops / Wireless devices and an internet cloud.
3. Connect the wireless Router to the Internet cloud using an ethernet cable.
4. Configure the SSID of the wireless router.
5. Enable the DHCP server on the wireless router
6. Configure the DHCP IP address range
7. Connect Laptops to the wireless network.
8. Set the Laptops to obtain IP address automatically using DHCP.
9. Check whether the devices receive IP address automatically
10. Use the ping command to test connectivity.
11. Use Simulation mode to observe packet transmission

RESULT:

The wireless internetwork was successfully configured using a wireless router, DHCP server and Internet cloud. IP address were automatically assigned to wireless devices and connectivity were successfully verified


Cloud-PT
Cloud1
WRT300N
Wireless Router0
PC-PT
PC0
Laptop-PT
Laptop0
Laptop-PT
Laptop1

10.0 Internetwork using switch, Router and Ethernet Cables.

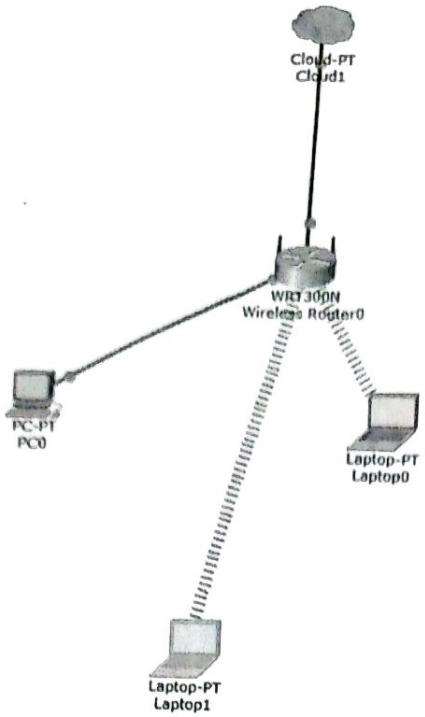
AIM: To design and configure an internetworking using switches, a router and Ethernet cables and to establish communication between different networks.

STEPS:

1. Open Cisco packet Tracer
2. Add 1 router, 2 switches and 3-4 PCs
3. Connect the PCs to the switches using Ethernet cables.
4. Connect the switches to the router using Ethernet cables.
5. Assign suitable static IP addresses to the PCs.
6. Configure the IP addresses of the router interfaces.
7. Enable the router interface using no shutdown.
8. Configure the appropriate default gateway on each PC
9. Use the ping command to test communication between PCs
10. Select Simulation Mode and observe the packet flow through the switches and router

RESULT:

The internetwork was successfully designed and configured using switches and router and Ethernet cables. Communication between the different network was established and verified successfully using the ping command.




## Command Prompt

```
Packet Tracer PC Command Line 1.0
PC>ping 192.168.30.1

Pinging 192.168.30.1 with 32 bytes of data:

Reply from 192.168.30.1: bytes=32 time=0ms TTL=255
Reply from 192.168.30.1: bytes=32 time=1ms TTL=255
Reply from 192.168.30.1: bytes=32 time=0ms TTL=255
Reply from 192.168.30.1: bytes=32 time=0ms TTL=255

Ping statistics for 192.168.30.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

PC>
```

## Command Prompt

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PC>
```